CYCLE 4

Secondary One - Term 1

Unit 1: Introduction to Geography **Duration: 27 periods of 40 minutes [8 weeks]**

Introduction:

In this unit, the new entrants to the secondary education system will be provided with a thorough understanding of what constitute geography as a subject. Students will be introduced to the main themes, concepts and issues that are central to the study of geography. The unit should help create the awareness that the study of geography is based on a number of fundamental intertwined concepts: location, pattern, processes, spatial interaction and human interaction with the environment. The importance of mapskills and graphicacy (the use of models, graphs, photographs, maps and other visual material) will also be stressed. Through this unit, the students will also be made aware of the significance and the contribution that the study of geography makes to an individual's education. In the last part of the unit the emphasis will be on the Earth's physical characteristics.

Terminal objectives:

1. Understand that geography is the study of the earth (the physical environment), its inhabitants (the human environment) and the inter-relationship between them.
2. Understand that geographical studies are based on a number of fundamental, intertwined concepts: location (where things are found), pattern (how they are distributed), spatial interaction (how do these things relate to each other), process (what are the processes that produce these things), and human interaction with the environment.
3. Recognise that geography is also concerned with the study of environmental, economic, cultural, political and resources-related issues (pollution and conservation, over-population, resource management, migration and refugees, political conflicts, etc.)
4. Recognise that geography draws upon a wide range of other disciplines and sciences: history, economics, geology, chemistry, biology, etc.
5. Know that geography is studied under different human and physical themes.
6. Recognise graphicacy and mapskills as important aspects of geography.
7. Understand the importance of the study of geography to their overall education: to providing them with a sense of their own locality and that of the world around them.
8. Know the main characteristics and features of the earth.
9. Understand the main movements of the earth and effects they produce.
10. Know the four main phases of the moon and the effects it produces on earth

Unit Assessment:

The terminal objectives of this unit will be assessed and graded based on:

(i) One written assessment at the end of the unit based on terminal objectives 1 to 10.
(ii) Different pieces of written work done during the course of the unit.
(iii) Informal oral assessments based on knowledge, cognitive skills, and attitudes.

Geography Teaching/Learning Scheme

Unit 1: Introduction to Geography.

Enabling objectives	Teaching/Learning Strategies	Resources	Assessment
• Explain that geography is the study of the earth and its inhabitants. • Identify and define the three branches (themes) of Geography. • Give clear examples of the three branches. • Explain the inter-relationship between them. • List the disciplines that geography draws from. • Explain the importance of obtaining a sound geographical education. • Identify local or global issues that are of importance to geography. • Explain why it is important to study geography in terms of location, patterns, spatial interaction, processes and human interaction with the environment. (Geographical systems and Ecosystems). • Identify the different types of illustrations used in Geography. • Explain the significance of map skills: - Define maps. - List the 5 main features of a map. - Name two types of maps: Political and Physical. - Describe the two types of maps. • Explain the significance of graphicacy: - Name the three main types of graphs (Bar, Line, Pie chart).	• Introduction using pictures and photographs • Photograph/picture analysis • Teacher expositions and explanation • Oral questioning • Discussion of issues using relevant sources (from textbooks, newspapers, magazines and other documents) • Written activities. • Map interpretation.	• Textbook: 1. Steps in Geography. 2. New Key Foundations 3. KS3 • Relevant handouts. • Relevant documents: newspapers, magazines. • Basic map work and photography. • Globe. • Wall maps. • Atlas • Blank maps of the world.	• Structured source-based written questions. • Informal assessments based on oral questioning and discussions.

CYCLE 4

Secondary One – Term 1

Unit 2: The Earth in the Solar System. **Duration: 12 periods of 40 minutes [4 weeks]**

Introduction:

The unit will look at the main characteristics of the Earth, which as far as scientists know, is the only planet in the universe with life. The students will be made to understand that we are an integral part of the universe and that other bodies such as the Sun and Moon have an influence on life on Earth. In the last part of the unit the emphasis will be on the Earth's physical characteristics.

Terminal objectives:

1. Know the main characteristics and features of the earth.
2. Understand the main movements of the earth and effects they produce.
3. Know the four main phases of the moon and the effects it produces on earth.

Unit Assessment:

The terminal objectives of this unit will be assessed and graded based on:

(i) One written assessment at the end of the unit based on terminal objectives 1 to 3.
(ii) Different pieces of written work done during the course of the unit.
(iii) Informal oral assessments based on knowledge, cognitive skills and attitudes.

Geography Teaching/Learning Scheme

Enabling objectives	Teaching / Learning Strategies	Resources	Assessment
• Describe the shape of the earth. • Illustrate that the earth has an axis, which is tilted. • State the difference between rotation and revolution. • Demonstrate rotation and revolution with the help of objects and charts. • Describe how rotation causes night and day, and how revolution causes the seasons. • State that the moon is the earth's satellite • State the four main phases of the moon. • Describe the effects of the phases of the moon to life on earth. • Name the two latitudinal hemispheres. • Name and locate the main lines of latitude on a world map. • Name the two longitudinal hemispheres. • Name and locate the main lines of longitude on a world map. • Use lines of latitude and longitude to locate places on a world map. • Name and locate some of the world's main physical features such as oceans, continents, mountains, ranges, deserts, seas, lakes, rivers, bays, gulfs. • Name and locate some of the major cities / capital cities in each continent (and the countries in which they are found).	• Brainstorming activities. • Explanation and expositions by teacher. • Oral questioning. • Discussions. • Demonstration of the concepts of rotation and revolution, and their effects using objects. • Show the axis and tilt of the earth using a globe. • Practical map-work exercises: the main lines of latitude and longitude. • Atlas and map-work activities: locate the main physical features on a world map.	• The Globe. • Charts of the Solar System. • New Key Foundations. • Atlas. • Blank World Maps. • Introduction to Geography handbook. • Relevant handouts. - Sun, Moon and Stars. - General Physical Geography (Bunnett). • Videos	• Structured source-based written questions. • Informal assessments based on oral questioning and discussions.

CYCLE 4
Secondary One - Term 2

Duration: 24 periods of 40 minutes [8 weeks]

Unit 3: Weather

Introduction:

In this unit students will build on the concept of weather. They will review their knowledge of the various elements of weather as well as the different instruments that are used to measure them. Students should be able to label, read, collect and interpret the measurements of these instruments. Students should know where they are located and why.

Terminal objectives:

1. Know the various weather elements. (rainfall, temperature, sunshine, wind speed and direction, humidity, cloud cover, pressure,)
2. Know what instruments are used to measure the elements.
3. Know the layout of the weather station.

Unit Assessment:

The terminal objectives of this unit will be assessed and graded based on:

(iv) One written assessment at the end of the unit based on terminal objectives 1 to 4.
(v) Different pieces of written work done during the course of the unit.
(vi) Informal oral assessments based on knowledge, cognitive skills and attitudes.

Enabling objectives	Teaching/Learning Strategies	Resources	Assessment
• State the location of each of the instruments in the weather station (layout). • Describe the features of the Stevenson Screen. • Describe the characteristics of the different weather instruments (Rain gauge, thermometers: Six's and Minimum and Maximum, Hygrometer, Barometer, Sunshine, Wind Vane, Anemometer). • Describe how the instruments mentioned above are used to measure the weather elements. • Make calculations using information from weather instruments : - Mean, diurnal ranges. • Identify the main types and amount of clouds. • Describe the characteristics of the different types of clouds. • List the three different types of rainfall: Convectional, Relief and frontal. • Describe the formation of the three types of rainfall mentioned above.	• Brainstorming activities. • Explanation and expositions by teacher. • Oral questioning. • Discussions. • Group-work. • Written activities. • Fieldwork: weather investigation. • Visit: meteorological station at the Seychelles International airport. • Map-work: interpreting isohyet maps of Seychelles. • Relevant documentaries on video.	• Weather handbook. • Textbooks: 1. Discover Physical and Environmental Geography 2. Geography Complete IGCSE. 3. Geography IGCSE Coursebook. 4. KS3. • Website: Meteorological Station Seychelles • Model of instruments (if available) • Videos (if available) • Simple weather maps.	• Structured source-based written questions. • Informal assessments based on oral questioning and discussions.

Review the following:
Definition of weather and climate. Weather instruments.

CYCLE 4

Secondary One - Term 2

Unit 4: Introduction to Climate Change **Duration: 15 periods of 40 minutes [4 weeks]**

Introduction.

In this unit students will develop knowledge on the causes and effects of climate change. The different solutions to climate change both local and global. Seychelles will be used as a case study example.

Terminal objectives:

1. Understand what climate change is.
2. Know the main causes of climate change.
3. Recognize the different effects of climate change.
4. Suggest possible solutions for dealing with effects of climate change.

Unit Assessment:

The terminal objectives of this unit will be assessed and graded based on:

(i) One written assessment at the end of the unit based on terminal objectives 1 to 5.
(ii) Various pieces of written work done during the course of the unit.
(iii) Informal oral assessments based on knowledge, cognitive skills and attitudes.

Enabling Objectives	Teaching/ Learning Strategies	Resources	Assessment
• Define Climate Change. • What are the causes of Climate Change? - Human activities: Pollution (greenhouse gas emissions), deforestation, global warming. • What are the effects of climate change? - Increase in global temperature, change in rainfall pattern (drought and flooding), sea level rise, acidification, desertification. • Suggest possible solutions to deal with issues of climate change. - Policy and Planning, Mitigation green house gases Adaptation techniques – Ecosystem based adaptation, Rehabilitation, Reafforestation, Renewable Energy • Case Study example: Climate Change Seychelles.	• Explanation and expositions by teacher. • Oral questioning. • Discussions. • Group-work. • Written activities and exercises. • Interview key stakeholders: fishermen, residents on coast.	• Blue Economy handbook. • Videos- Effects of climate change in Seychelles. • Textbook on climate change MOE. • Field experience –Coastal : Anse Kerlan, Anse La Mouche, Anse Boileau, North East Point, Bel Ombre, Beau Vallon, La Digue. • Baie Lazare- Wetlands Rehabilitation.	• Structured source-based written questions. • Informal assessments based on oral questioning and discussions

CYCLE 4

Secondary One - Term 3-

Unit 5: Introduction to Population Studies **Duration: 15 periods of 40 minutes [5 weeks]**

Introduction:

This is an introductory unit of population studies, a topic that will be revisited in more depth in cycle 5. In this introduction, students will get the chance to become familiar with some of the main concepts of population geography such as population growth and the factors that affect it, the pyramid and its features, and population density and distribution. Aspects of the Seychelles demography should be used as much as possible as examples, but it is expected that other relevant examples from around the world can be used.

Terminal objectives:

1. Understand the terminologies associated with population studies.
2. Understand the trends in population growth.
3. Know the factors that affect the population of a country.
4. Recognise the chief features of a population pyramid.
5. Know the factors that affect density and distribution of population.

Unit Assessment:

The terminal objectives of this unit will be assessed and graded based on:

(i) One written assessment at the end of the unit based on terminal objectives 1 to 5.
(ii) Various pieces of written work done during the course of the unit.
(iii) Informal oral assessments based on knowledge, cognitive skills and attitudes.

Geography Teaching/Learning Scheme

Unit 5: Introduction to Population Studies

Enabling objectives	Teaching/Learning Strategies	Resources	Assessment
• Define population and terms associated with population: -Birth rate, Death rate, Life expectancy, natural increase/ decrease, infant mortality, literacy rate. • Describe the trend in the world's population growth. • State that the world's population is not evenly distributed. • Define the following terms: distribution of population and density of population • List the factors that affect the distribution of population and density of population. • Explain briefly the factors that affect the density and distribution of population. • Identify the main areas of high and low population densities on a distribution map of the world. • State the factors that affect the population of a country. • List the factors which influence the birth and death rate of a country • List some factors that affect life expectancy. • State what is a population pyramid. • Identify the main features of a population pyramid (age, sex, birth rate, death rate, life expectancy). • Describe the population pyramid of Seychelles.	• Explanation and expositions by teacher. • Oral questioning. • Discussions and debates. • Group-work. • Written activities and exercises using simple population statistics. • Analysis of simple graphs showing population growth, birth rate and death rate. • Interpretation of a typical population pyramid. • Interpretation of population pyramid for Seychelles. • Analysis of simple population density and distribution maps. • Relevant documentaries on video.	• Textbook: 1. Geography in Place 2 2. The New Wider World. 3. KS3. 4. IGCSE Collins. • Maps: - Density and Distribution maps of Seychelles Population pyramid- Seychelles. • Videos (if available)	• Structured source-based written questions. • Informal assessments based on oral questioning and discussions.

Geography Teaching/Learning Scheme **Duration: 12 periods of 40 minutes [4 weeks]**

Unit 6: Introduction to Settlement Studies

Introduction:

In this unit the concept of settlements will be introduced, where pupils will be exposed to some of the basic theories in the topic. The different types of settlements that exist and the different methods in which they are classified: according to size, shape, range of services offered, etc.; will form the bulk of the unit. A study of the internal layout of bigger settlements such as towns and cities will also be introduced in relation to the theoretical concentric model of a city (after Burgess). These will be studied especially with reference to Seychelles as far as possible. This topic will be revisited in cycle 5 where a more in-depth study will be carried out.

Terminal objectives:

1. Distinguish between villages, towns and cities as types of settlement.
2. Understand that settlements are classified according to their sizes and the range of services they offer.
3. Distinguish between rural and urban settlements.
4. Know the main types of settlement patterns in Seychelles: linear, nucleated and dispersed.
5. Understand the functional zones in a city according to the Burgess concentric model.

Unit Assessment:

The terminal objectives of this unit will be assessed and graded based on:

(iv) One written assessment at the end of the unit based on terminal objectives 1 to 5.
(v) Various pieces of written work done during the course of the unit.
(vi) Informal oral assessments based on knowledge, cognitive skills and attitudes.

Enabling objectives	Teaching/Learning Strategies	Resources	Assessment
• Define settlements. • Define site and situation. • Differentiate between a rural and an urban settlement. • Give examples of rural and urban settlements in Seychelles. • List the functions of a rural and an urban settlement. • Define the term settlement hierarchy. • Classify settlements according to their sizes and range of services. • List the types of services offered by each type of settlement. • Draw and label the concentric model of urban settlements. • List the main activities that occur in each zone in the model. • Name some important characteristics of a CBD. • Identify and illustrate the three main settlement patterns: linear, nucleated and dispersed. • List simple reasons why such patterns develop (in the local context). • Give examples of each type of settlement patterns in Seychelles.	• Explanation and expositions by teacher. • Oral questioning. • Discussions. • Group-work. • Written activities and exercises. • Fieldwork: survey of types of services found in the local area/district (and Victoria if possible). • Interpretation of the concentric model of urban areas: draw and label the model. • Map-work: identification of settlement patterns.	• Textbooks: 1. The New Wider World 2. Discover Human Geography 3. Geography in Place 1 4. New Key Foundation. • Maps and photographs of settlement patterns.	• Structured source-based written questions. • Informal assessments based on oral questioning and discussions.

CYCLE 4

Secondary One - Term 3

Unit 7: Tourism: Trends and Impacts **Duration: 9 periods of 40 minutes [3 weeks]**

Introduction:

The introduction of this unit on Tourism to S1 students is based on the fact that this is a topic with which our students are relatively familiar. The aim is to introduce them to the concept of tourism, and basic knowledge about how the industry works and focus their attention on the importance of tourism to developing countries like Seychelles. The topic is developed in a way that enables pupils to base knowledge on their experiences from the local environment as well as gain insight and knowledge about the experiences of other countries. Therefore case studies from around the world should be used but at this stage special emphasis should be given to the development of tourism in Seychelles.

Terminal objectives:

1. Understand the term tourism.
2. Recognise tourism as one of the most rapidly expanding industries in the world.
3. Know the factors leading to the world-wide growth of tourism.
4. Know the difference between domestic and international tourism.
5. Know the origins of tourists that come to Seychelles.
6. Understand how the tourism industry is organised in Seychelles.
7. Know some of the major tourist attractions of the world
8. Know that the tourism industry is an important part of the development strategy for developing countries.
9. Know the advantages and disadvantages of tourism.

Unit Assessment:

The terminal objectives of this unit will be assessed and graded based on:

(i) One written assessment based on objectives 1 to 9.
(ii) Different pieces of work done during the course of the unit.
(iii) Informal assessment based on knowledge, cognitive skills and attitudes.

Geography Teaching/Learning Scheme

Unit 7: Tourism; Trends and Impacts

Enabling objectives	Teaching/Learning strategies	Resources	Assessment
• Define the term tourism. • Define domestic and international tourism. • Mention that tourism is one of the world's fastest growing industries. • State that global tourism really flourished after the1950s. • Give reasons for the rapid recent growth of worldwide tourism. • Explain how the tourism industry is organised in Seychelles. • List the origins of tourists that visit Seychelles. • Describe the main tourist attractions in Seychelles in terms of climate, scenery, ecology and culture. • Name some of the famous tourist attractions/destinations in other parts of the world. Explain briefly why these destinations are popular. • Explain the importance of the tourism industry to the Seychelles economy. • Explain the importance of tourism to developing nations. • Describe some of the main positive and negative impacts of tourism.	• Discussions. • Written activities. • Group work. • Research/ Project work. • Interpretation of graphs, pie charts, etc. • Activities centered on the use of tourism statistics. • Field work e.g. a brief study of tourism infrastructure/land-use in a small area; a simple tourism survey, for example at the airport; a simple study of some impacts of tourism.	• The New Wider World. • Geography in Place 2 • Website: STB. • Brochures and other documents available at the documentation centre at the Ministry of Tourism. • Tourism statistics from NBS office.	• Structured source-based written questions. • At least one project-type assignment. • Informal assessments based on oral questioning and discussions.

CYCLE 4

Secondary Two – Term 1

Unit 8: Introduction to Contour Maps: Physical and Human Features. **Duration: 21 periods of 40 minutes [6 weeks]**

Introduction:

Since physical and human features on the earth's surface are frequently represented on maps, geography students need to develop the skills to read and interpret maps, if they are to become competent geographers. Bearing in mind the difficulty some pupils encounter when doing map-work for the first time, the units on map-work have been developed in a number of stages. In this first unit the student is introduced to contour maps for the first time, and the emphasis will be on recognising and identifying physical features only on maps. Subsequent units will build up on this knowledge. Local Ordnance Survey (O.S.) maps, as well as relevant topographic maps may be used.

Terminal objectives:

1. Know the main features found on contour maps.
2. Understand the methods of showing heights on maps.
3. Recognise topography indicated by contours maps.
4. Recognise landforms produced by contour patterns.
5. Recognise different human features shown on maps.
6. Show an understanding of how site influences the development of different types of settlements patterns.
7. Show an understanding of how situation influences the development of different types of settlements patterns.
8. Recognise how settlements are shown on maps.
9. Recognise different forms of communication on maps and how they are influenced by physical landscapes.
10. Recognise different land-uses shown on a map.

Unit Assessment:

The terminal objectives of this unit will be assessed and graded based on:

(i) One written assessment based on objectives 1 to 10.
(ii) Activities and exercises done during the course of the unit.
(iii) Informal assessment based on the acquisition of the skills involved.

Geography Teaching/Learning Scheme

Unit 8: Introduction to Contour Maps: Physical Features and Human Features.

Enabling objectives	Teaching/Learning strategies	Resources	Assessment
• Name the three types of scale. • Explain the importance and uses of the scale. • Explain the importance and uses of the key (Conventional signs). • List and describe the methods of showing heights on maps (use of colours, contours, spot heights, trigonometrical station). • Calculate Vertical Interval. • State that contour patterns describe landforms on the ground. • Identify the different types of slopes indicated by contour patterns (uniform, gentle, steep). • Differentiate between a hill and a mountain. • Identify hills and mountains on a contour map. • Differentiate between a valley and a spur. • Identify valleys and spurs on a contour map. • Define a headland and a bay. • Identify headlands and bays on a contour map. • Differentiate between a plain and a plateau • Locate plains and plateaux on a contour map. • State what a ridge is. • Locate ridges on a contour map. Human features.	• Teacher exposition and explanations • Group work • Map interpretation • Field exercises • Observation	• Reading Tropical Maps. • Basic Map-work and Photography. • Map Reading: Geography Book 3/4. • O.S. Maps of Seychelles.	• Structured questions based on maps. • Informal assessments based on classroom performance.

• Identify the human features shown on maps (houses, building, roads, schools, churches, etc) with the use of the key. • Use illustrations to show how some of these features are shown on maps. • Identify how settlements are shown on maps • Describe the site and situation of a variety of settlements on any given map. • Identify the three types of settlement patterns on maps. • Identify the forms of communication on maps (footpath, track, minor road, main road, etc) with the use of the key. • Illustrate how these forms of communication are shown on maps. • Explain how the physical landscape influence communication patterns on maps. • Identify the land-use pattern on maps (natural vegetation, agriculture, mining, fishing, industrial, commercial, recreational, and administrative) with the use of the key. • Explain the influence of physical landscapes on land-use patterns on maps			

CYCLE 4

Secondary Two - Term 1

Unit 9: The structure of the earth **Duration: 12 periods of 40 minutes [3 weeks]**

Introduction:

Having gained a basic understanding of rock types and formation in the previous unit, the students are now ready to study the characteristics and composition of the Earth. This will form the basis upon which the theory of Plate Tectonics will be introduced. It will also help explain the formation of some of the major surface features on Earth, such as fold-mountains, volcanoes, mid-oceanic ridges, ocean trenches and island arcs. To link all of this to our local context, the last part of the unit involves a brief look at how the formation of the Seychelles islands could be linked to these theories.

Terminal objectives:

1. Know the main parts of the earth.
2. Know the main characteristics of each layer of the earth.
3. Understand the difference between continental and oceanic crusts.
4. Recognise the world's major plates.

Unit Assessment:

The terminal objectives of this unit will be assessed and graded based on:

(i) One written assessment based on objectives 1 to 4.
(ii) Different pieces of work done during the course of the unit.
(iii) Informal assessment based on knowledge, cognitive skills and attitudes.

Geography Teaching/Learning Scheme

Unit 09: The structure of the earth.

Enabling objectives	Teaching/Learning Strategies	Resources	Assessment
• Use diagrams to identify the main parts of the earth (inner core, outer core, mantle, crust). • Compare the relative thickness of the layers of the earth. • Describe the main characteristics of each layer; their state and temperature. • State that the crust of the earth is made up of plates which can fit like a jigsaw puzzle. • State that plates are movable floating layers. • Differentiate between oceanic and continental crusts/plates. • Name and locate the major plates that make up the earth's crust.	• Discussions. • Written activities. • Drawing activities. • Group work. • Research work. • Practical activities: making a model of a globe showing the major plates. • Demonstrations e.g. of the movements at different boundaries; of the formation of fold-mountains.	• Discover Physical Geography. • The New Wider World. • Geography in Place 1. • Physical Geography in diagrams for Africa. • KS3	• Structured source-based written questions. • Informal assessments based on classroom performance.

CYCLE 4

Secondary Two - Term 1

Unit 10: Rocks and Minerals **Duration: 6 periods of 40 minutes [3 weeks]**

Introduction:

This unit will give S2 student a chance to develop an awareness of the world in which we live by studying the nature of the materials that make up the Earth. The pupil should be made to understand that the term 'rocks' refers to the whole range of materials that make up the Earth, from the hardest granites to the softest clays. They should also understand that rocks originate in different ways and this is what differentiates them from one another. Finally the value of rocks to humans should be touched upon.

Terminal objectives:

1. Know what rocks and minerals are.
2. Understand the terms used to describe rocks.
3. Know that rocks are the materials that make up the composition of the Earth's surface.
4. Recognise the types of minerals found in rocks.
5. Understand the formation of igneous, sedimentary and metamorphic rocks, and that they originate in different ways.
6. Know examples of each of these types of rock.
7. Show an understanding of the importance of rocks to humans.
8. Know the types of rocks found in Seychelles.

Unit Assessment:

The terminal objectives of this unit will be assessed and graded based on:

(i) One written assessment based on objectives 1 to 8.
(ii) Different pieces of work done during the course of the unit.
(iii) Informal assessment based on knowledge, cognitive skills and attitudes.

Geography Teaching/Learning Scheme

Unit 10: Rocks and minerals

Enabling objectives	Teaching/Learning Strategies	Resources	Assessment
• Define rocks (as materials that constitute the crust of the Earth). • Define the term minerals. • List the types of minerals found in rocks. • Describe the characteristics of rocks (hardness, permeability, resistance). • Name the three main types of rocks, giving examples for each type. • Explain the formation of igneous rocks with examples. • Explain the formation of sedimentary rocks with examples. • Explain the formation of metamorphic rocks with examples. • List the importance of rocks to humans. • Identify the types of rocks that are common in Seychelles.	• Discussions. • Written activities. • Group work. • Research activities. • Practical activities: e.g. finding rock samples from the local area. • Visit to the Natural History Museum (exhibits of types of rocks found in Seychelles). • Visit to Gondwana Granite at Providence. • Visit to local area showcasing examples for uses of rocks eg. Cemetery, town, coastal areas.	• The New Wider World • Physical Geography in diagrams for Africa. • Videos. • Samples of rocks. • Pictures showcasing uses of rocks: Eg. Rock armouring, Tombstone, Buildings, Monuments.	• Structured source-based written questions. • Informal assessments based on classroom performance.

CYCLE 4

Secondary Two - Term 2

Unit 11: Plate Tectonics. **Duration: 15 periods of 40 minutes [5 weeks]**

Introduction:

Having gained a basic understanding of rock types and formation in the previous unit, the students are now ready to study the characteristics and composition of the Earth. This will form the basis upon which the theory of Plate Tectonics will be introduced. It will also help explain the formation of some of the major surface features on Earth, such as fold-mountains, volcanoes, mid-oceanic ridges, ocean trenches and island arcs. Gain a basic understanding of the theory of Continental Drift put forward by Alfred Wegener in 1912.

1. Realise how this theory was later revised to form the new theory of Plate Tectonics.
2. Recognise and understand the major plate movements and associated features.
3. Understand how the formation of the islands of Seychelles is related to the theory of drifting continents and plate tectonics.

Enabling objectives	Teaching/Learning Strategies	Resources	Assessment
• Describe the theory of Continental Drift. • Explain how this theory was revised to form the new theory of Plate Tectonics. • Define the term plate tectonics. • State that tension forces cause plates to move apart and the compression forces cause them to move towards each other. • Define compression forces. • Define Tension forces • Name the three types of plate margins and the movement associated with each one. • Demonstrate the three types of movements. • Identify the resulting landforms and features associated with each one.	• Discussions. • Written activities. • Drawing activities. • Group work. • Research work. • Demonstrations e.g. of the movements at different boundaries; of the formation of fold-mountains.	• Discover Physical Geography. • The New Wider World. • Geography in Place 1 • Physical Geography in diagrams for Africa. • Videos. • Complete IGCSE. • KS3	• Structured source-based written questions. • Informal assessments based on classroom performance.

CYCLE 4

Secondary Two - Term 2

Unit 12: Folding and Faulting **Duration: 9 periods of 40 minutes [2 weeks]**

Introduction:

Having gained a basic understanding of the theory of Plate Tectonics, and formation of rocks the students will now be introduced to internal forces which produced landforms. This unit will built students knowledge on Vertical and Lateral Earth movement resulting in the formation of features such as block mountains, fold mountains, rift valleys, basics escarpments.

Terminal objectives:

5. Understand the internal forces which produced landforms.
6. Recognise that vertical movements causes crustal rocks to fault.
7. Recognise that lateral movement cause crustal rocks to fold.
8. Knows the features produced by both lateral and vertical movements.

Unit Assessment:

The terminal objectives of this unit will be assessed and graded based on:

(i) One written assessment based on objectives 1 to 4.
(ii) Different pieces of work done during the course of this unit.
(iii) Informal assessment based on knowledge, cognitive skills and attitudes.

Unit 12: Folding and Faulting

Enabling objectives	Teaching/Learning strategies	Resources	Assessment
• Define folding. • Draw the main types of folds. • Demonstrate the formation of fold mountain and give examples. • Locate fold mountains on a world map. • Define faulting. • List and draw the different types of faults giving examples. • Use diagrams to describe the different features formed by faulting (block mountains / rift valleys) • Locate block mountains and rift valleys on a world map.	• Demonstration using plasticine. • Group work • Group discussion • Drawing of folds, faults • Teacher's explanation • Pictures	• Text book: Physical Geography in Diagram for Africa. • The New Wider World. • Discover Physical Geography. • Diagrams on folding and faulting.	• Structured source based questions. • Informal assessments based on classroom performance.

Review tension and compression forces to begin topic.

CYCLE 4

Secondary Two - Term 2

Unit 13: Introducing volcanoes. **Duration: 15 periods of 40 minutes [5 weeks]**

Introduction:

This unit is an extension of the unit 12 and attempts to develop the knowledge of the pupils in understanding the range of volcanic and seismic activities prevalent on the surface of the earth. It also attempts to make pupils realise the effects these activities have on our lives. While these activities will be studied on a global scale, examples from the region have been used as far as possible.

Terminal objectives:

1. Know what volcanic activities are.
2. Understand the location of volcanoes and earthquakes on the surface of the earth.
3. Understand the different types of materials emitted from volcanoes and how they determine the types of volcanoes that occur.
4. Know the extrusive volcanic features and some of the intrusive features that occur.
5. Understand that intrusive volcanic features only form part of the landscape after exposure.
6. Know the structure of a volcano.
7. Know the different types of volcanoes.
8. Know the stages of a volcano.
9. Know the types of volcanic features that occur in our region.
10. Know how and why earthquakes occur, using examples.
11. Recognise that volcanic and seismic activities have both advantages and disadvantages to humans.

Unit Assessment:

The terminal objectives of this unit will be assessed and graded based on:

(i) One written assessment based on objectives 1 to 12.
(ii) Different pieces of work done during the course of the unit.
(iii) Informal assessment based on knowledge, cognitive skills and attitudes.

Geography Teaching/Learning Scheme

Unit 13: Introducing volcanoes.

Enabling objectives	Teaching/Learning Strategies	Resources	Assessments
• Define volcanic activity. • Locate and describe the distribution of volcanic and seismic (earthquake) activities on the earth's surface. • Explain why volcanic and seismic activities occur at plate boundaries. • Differentiate between intrusive and extrusive volcanic features. • List all the examples of extrusive and intrusive volcanic features (volcanoes, lava plateaux, hot springs and geysers). (Sills, dykes, batholiths, laccoliths only). • State that it is only after exposure that intrusive features form part of the landscape. • Describe the structure of a volcano. • Give examples of the materials emitted. • Differentiate between magma and lava • Name and describe the three stages in the life cycle of a volcano, giving examples from around the world. • Name and give examples of different types of volcanoes : Shield and Composite (Strato). • List the advantages and disadvantages of volcanic eruptions to humans, especially those who live in their vicinity. • Case Study: Mt. Etna as example. - Cause - Effects - Impacts - Prediction and Precaution	• Discussions. • Written activities. • Drawing activities. • Group work. • Map exercises. • Project work / making models.	• The New Wider World. • Geography in Place 1. • Physical Geography in diagrams for Africa. • Complete Geography IGCSE. • KS3. • World map showing distribution.	• Structured source-based written questions. • Informal assessments based on classroom performance.

CYCLE 4

Secondary Two - Term 3

Unit 14: Introducing earthquakes. **Duration: 15 periods of 40 minutes [3 weeks]**

Introduction:

This unit is an extension of the unit 13 and attempts to develop the knowledge of the pupils in understanding the range of seismic activities prevalent on the surface of the earth. It also attempts to make pupils realise the effects these activities have on our lives. While these activities will be studied on a global scale, examples from the region have been used as far as possible.

Terminal objectives:

12. Know how and why earthquakes occur, using examples.
13. Recognise that seismic activities have both advantages and disadvantages to humans.

Unit Assessment:

The terminal objectives of this unit will be assessed and graded based on:

(i) One written assessment based on objectives 1 to 12.
(ii) Different pieces of work done during the course of the unit.
(iii) Informal assessment based on knowledge, cognitive skills and attitudes.

• Define earthquake. • Explain the characteristics of earthquakes (focus, epicentre). • Discuss the methods of monitoring and detecting earthquakes. • List the advantages and disadvantages of seismic activities to humans, especially those who live in their vicinity. • State why Seychelles does not experience any earthquake. • Compare and contrast effects of earthquakes between MEDCS and LEDCS. • Case Study Tsunami Sumatra, Indonesia 2004. - Causes - Effects - Impacts - Prediction and Precaution.	• Discussions. • Written activities. • Drawing activities. • Group work. • Project work / making models.	• The New Wider World. • Geography in Place 1. • Physical Geography in diagrams for Africa. • Complete Geography IGCSE. • KS3. • World map showing distribution.	

CYCLE 4

Secondary Two - Term 3

Unit 15: Natural Regions (Tropical Ecosystems) **Duration: 09 weeks**

Introduction:

It is a widely accepted fact that the environment of Seychelles is a vital concern for every Seychellois, therefore in this unit the student is given the opportunity to develop insight and awareness of their immediate and regional environments, and the place that Man occupies in them. For this cycle the unit will focus on two major types of tropical eco-systems only: Tropical Rainforests and Hot Deserts types of ecosystems that our pupils are more familiar with. The concept of the ecosystem will be introduced: in each case, the student should be engaged in a study of the main features of the eco-system itself, and then look at the impact of human activities on these eco-systems and the ways in which these can be minimised. Therefore the student is made to realise that the eco-system in which they live should be carefully nurtured and managed for increased sustainability.

Terminal objectives:

1. Understand what natural regions are.
2. Know the main features of the types of natural regions listed: a) tropical rainforests and b) hot deserts.
3. Recognise the different climatic features of each of these regions.
4. For each region show an understanding of how animals and plants have adapted to the climate.
5. Recognise human activities taking place in each region and their impacts.
6. Understand that careful management of our ecosystem can help reduce the negative impacts.

Unit Assessment:

The terminal objectives of this unit will be assessed and graded based on:

(i) Two written assessments based on objectives 1 to 6. One assessment should be based on the rain forest ecosystem or hot desert regions.
(ii) Different pieces of work done during the course of the unit.

(iii) Informal assessment based on knowledge, cognitive skills and attitudes.

Geography Teaching/Learning Scheme

Unit 15: Tropical Ecosystems

Enabling objectives	Teaching/Learning Strategies	Resources	Assessment
• Define natural regions. • Describe factors influencing climate (latitude, altitude, relief, ocean currents, and prevailing winds). • List the examples of and locate on a map natural regions across the globe (Tropical Rainforest, Tropical Monsoon, Savanan Grassland, Hot Deserts, Mediterenean, Tundra, Polar. Tropical rainforests • State where tropical rainforests are located, with examples. • Locate tropical rainforest regions on a map of the world: include Seychelles specific areas in Seychelles such as high altitude forest of Morne Blanc and Mont Dauban. • Describe the climate of the equatorial regions (rainfall and temperature). • Describe the structure of the tropical rainforest. (Shrub layer, under canopy, main canopy, emergents) • Describe the vegetation found in the rainforest. • Explain how the vegetation has adapted to the climate.	• Discussions. • Written activities. • Group work. • Research work. • Visit to the Natural History Museum (exhibits of types of forests found in Seychelles). • Field work: field excursions to study the trees of the Seychelles forest(s); Impact of human activities on ecosystems.	• The New Wider World. • Geography Complete IGCSE. • BBC Planet Earth Documentary. • BBC Planet Earth Rainforest. • IGCSE Collins.	• Structured source-based written questions • Informal assessments based on classroom performance.

• State what type of soil is found in the tropical rainforest. • Explain why soil in the tropical rainforest is poor or infertile. • Give examples of animals found in the rainforest. • Describe human activities taking place in the rainforest. Case Study: Borneo or Amazon - Describe the impacts of human activities on rainforest ecosystem. - Describe ways in which the negative impacts can be reduced. For Hot deserts: • State where they are located. • Locate them on a map of the world (showing their relative positions). • Describe the climate associated with each (the main features of rainfall and temperature). • Describe their vegetation. • Explain how the vegetation is adapted to the climate. • Give examples of animals found. Case Study: Namib or Arabian desert (Dubai). • Describe the type of human activities taking place. • Describe the impacts of human activity on the ecosystem. • Describe ways in which the negative impacts can be reduced. -			

CYCLE 4

Secondary Three - Term 1

Unit 16: Development and Trade **Duration: 9 periods of 40 minutes [4 weeks]**

Introduction:

In this unit the students will be introduced to the concept of development and trade. It should help develop students' understanding of world pattern of development and create awareness of the economic indicators used by the U.N. It will also help students to recognise the importance of sustainable development. This unit also look at the major trade groups of the world and their influence on a country's development.

Terminal objectives:

1. Know the terminologies associated with development.
2. Develop an understanding of the world pattern of development.
3. Understand why the UN uses GNP and HDI to measure development levels.
4. Know the factors that affect GNP and HDI.
5. Understand the differences in GNP between MEDCs and LEDCs.
6. Know the importance of NICs.
7. Show an understanding of the concept of sustainable development..
8. Know the major trade groups of the world.
9. Understand how countries benefit by trading together.
10. Understand how world trade works.
11. Know the role of the WTO to world trade.

Unit Assessment:

The terminal objectives of this unit will be assessed and graded based on:

(i) Two written assessments. One based on objectives 1 to 7, and one based on objectives 8 to 11.

(ii) Different pieces of work done during the course of the unit.
(iii) Informal assessment based on knowledge, cognitive skills and attitudes.

Geography Teaching/Learning Scheme

Unit 16: Development and Trade

Enabling objectives	Teaching/Learning Strategies	Resources	Assessment
• Define the terms Development, Globalization, MEDC, NIC, LEDC. • Describe how the influence can give an indication of the development of a country: GNP per capita, population growth, energy consumption, access to housing, health and education, employment / unemployment, nutrition and diet, agricultural production and levels of mechanisation, imports / exports, and communication. • Define the term GNP. • State that GNP and HDI are two common indicators of development. • State that a country's wealth is measured using GNP per capita figures. • State why GNP is given in US dollars. • Define the term HDI. • Explain why the HDI is preferable as a measure of development to GNP. • Compare and contrast the economic and social development of MEDCs and LEDCs. • List the countries that make up NICs (Newly Industrialised Countries and BRICS. • Define the term sustainable development.	• Discussions. • Written activities. • Group work. • Research activities. • Explanation • Drawing activities. • Project work • Group-work	• The New Wider World. • Geography in Place 1. • Geography in Place 2. • Discover Human Geography. • KS3 • Development and Trade handbook. • Geography IGCSE Coursebook Sibley.	• Structured source-based written questions. • Informal assessments based on classroom performance.

• Explain why sustainable development should be encouraged. Give examples: Seychelles. Project			

¥ Include Finance/ Economy somewhere.

• Define the term trade. • Give examples of trade groups, e.g. EU, AU, COMESA, SADC, OPEC, of which the EU is the largest trading group. • For each of the organisations above, name the member countries. • Discuss their main functions. • Name the organisations of which Seychelles is a member. • Describe the advantages and disadvantages of trading and trade groups to the member countries and to the world economy.			

• Describe the imbalance of trade between MEDCs and LEDCs. • Describe the role of the WTO. • State that the WTO is an organisation that supervises the implementation of trade agreement to settle trade disputes.			

CYCLE 4

Secondary Three - Term 1

Unit 17: Introduction to Industry and Fishing **Duration: 12 periods if 40 minutes [6 weeks]**

Introduction:

In this unit the students will be introduced with the term industry. It will develop an understanding of the different types of industry and the factors that influence their location. It will help students to recognise the importance of the Blue Economy and the fishing industry to the Seychelles economy. It will also create an awareness of the impact of industries on the environment.

Terminal objectives:

1. Understand the term industry.
2. Know the different types of industry.
3. Show an understanding of how the industrial system works.
4. Understand the factors that influence the location of industries.
5. Understand the importance of the Blue Economy to Seychelles.
6. Understand that fishing provide humans with a vital source of food.
7. Recognise the importance of the fishing industry to the Seychelles economy.
8. Know the different types of fishing carried out in Seychelles.
9. Understand that the physical / natural factors offer favourable conditions for the development of fishing in Seychelles.
10. Understand the impact of fishing on the environment.
11. Understand how secondary industries operate, using examples from the local area.
12. Understand the impact of primary and secondary industries on the environment.

Unit Assessment:

The terminal objectives of this unit will be assessed and graded based on:

(i) Two written assessments. One based on objectives 1 to 9, and one based on objectives 10 to 11.
(ii) Different pieces of work done during the course of the unit.
(iii) Informal assessment based on knowledge, cognitive skills and attitudes.

Geography Teaching/Learning Scheme

Unit 17: Introduction to Industry and Fishing

Enabling objectives	Teaching/Learning Strategies	Resources	Assessment
• Define the term industry. • State the meaning of primary, secondary, tertiary and quaternary industries. • Give examples of each of these types of industry. • Describe how an industrial system works (inputs, outputs and processes). • Name and explain the factors that determine the location of an industry. • Citing examples of each, distinguish between heavy and light industries. • Citing examples of each distinguish between fixed and footloose industries. • Using a local example of secondary industries (Case Study): - Describe the industry as a system. - Explain how the industry works. - Explain the factors that affect/determine its location. - State whether it is an example of a light/heavy and/or a footloose/fixed industry. - Describe its importance to the economic development of Seychelles. - Describe its impact on the environment. • Define Blue economy.	• Explanation • Discussion. • Drawing activities. • Project work • Group-work • Written activities. • Research work • Field **v**isits IOT/ Food pro/ Seybrew.	• The Human Geography of Seychelles. • Steps in Geography Book one. • The New Wider World. • Atlas for Seychelles. • Blue Economy handbook. • Field visit Aquaculture site at Providence.	• Structured source-based written questions. • Informal assessments based on classroom performance.

• Why is the Blue economy important for Seychelles? • Give examples of some key activities that form part of the blue economy e.g. aquaculture, Value added products e.g. Smoked fish etc. • State that fishing is an example of a primary industry. • Outline the importance of the fishing industry. • Identify the natural resources that favour the development of the fishing industry in Seychelles. • Locate on a map of Seychelles the main fishing grounds (banks) found within the Seychelles EEZ. • Differentiate between artisanal, semi-industrial and industrial fishing in Seychelles. • Describe the methods of artisanal fishing, semi-industrial and industrial fishing • For each type of fishing, name the types of fish caught.			

CYCLE 4

Secondary Three - Term 2

Unit 18: Map Skills: Ordnance Survey Maps

Duration: 12 periods of 40 minutes [4 weeks]

Introduction:

In this unit the students will build up on the knowledge gained in unit 7 and 10. With the basic understanding of human and physical features on contour maps, the student can now develop other skills required when dealing with topographic maps such as interpretation of features.

Terminal Objectives:

1. Know how to read large scale maps
2. Show an understanding on the use and analysis of O. S maps of Seychelles.
3. Develop the ability to give descriptions of landscape using geographical terms
4. Recognise differences in density of drainage, patterns of stream, and sizes of streams in relation to the relief.
5. Know how to identify and describe the land use variations in relation to the types of relief
6. Understand the land use patterns on O.S maps
7. Recognise the forms of communication shown on O.S maps
8. Develop an understanding of settlement patterns on O.S maps

Unit Assessment:

The terminal objectives of this unit will be assessed and graded based on:

(i) One written assessment based on objectives 1 to 5.
(ii) Different activities and exercises done during the courses of the unit.
(iii) Informal assessment based on knowledge, cognitive skills and attitudes.

Geography Teaching/Learning Scheme

Unit 18: Map Skills: Ordnance Survey Maps

Enabling objectives	Teaching/Learning Strategies	Resources	Assessment
• Difference between large / small scale map • Give direction on maps both as compass point and bearing • Calculate coordinate systems on maps using the reference system (four-figure and six-figure grid references) • Measure distances along straight and curved distances • Give brief descriptions of landscape using geographical terms (e.g. plains, gently-sloping land) • Describe the shape and form of river channel. • Draw cross sections • Describe settlement pattern on O.S maps. • Describe the land use patterns on O.S maps (e.g. recreational, tourism, agriculture, quarrying, fishing, commercial, industrial) • Read map extracts of Seychelles with a full key of the symbols used. • Interpret patterns of rivers in terms of relief. • Identify the forms of communication on maps.	• Explanation • Demonstration • Map interpretation	• Reading tropical maps • Map extract of Mahe, Praslin, La Digue • Maps from past IGCSE papers • New Key Foundation. • Map Skills Handbook.	• Assessment • Map work activity • Informal assessment based on map skills

CYCLE 4

Secondary Three – Term 2

Unit 19: Weathering and Mass Movements **Duration: 12 periods of 40 minutes [4 weeks]**

Introduction:

In this unit students will be introduce to the topic weathering and mass movement. It will enable student to recognise the importance of human actions in how they trigger and accelerate mass movement.

Terminal objectives:

1. Understand the difference between weathering and mass movements.
2. Know the main types of weathering.
3. Know the agents of weathering.
4. Know the factors which influence weathering.
5. Know the resulting features that are formed by the different types of weathering.
6. Know the different types of mass movements.
7. Understand the causes of mass movements.
8. Understand the effects of mass movements on the natural landscape and the threat they can pose to humans.
9. Understand how human activities may accelerate weathering and mass movements.

Unit Assessment:

The terminal objectives of this unit will be assessed and graded based on:

(i) Two written assessments: one based on objectives 1 to 5, and one based on objectives 6 and 9.
(ii) Different pieces of work done during the course of the unit.
(iii) Informal assessment based on knowledge, cognitive skills and attitudes.

Geography Teaching/Learning Scheme

Unit 19: Weathering and Mass Movements

Enabling objectives	Teaching/Learning Strategies	Resources	Assessment
• Define the terms weathering and mass movement, differentiating clearly between the two. • List the main types of weathering. • List the agents of weathering. • Describe the different processes involved in each type of weathering. • Explain how the following influence weathering: rock types, structure, resistance, permeability, climate, and time. • Describe the climatic conditions under which each type of weathering is more dominant. • State that physical weathering is more dominant in temperate latitudes and that chemical weathering is more dominant in tropical latitudes. • Name features resulting from the different types of weathering (granite). • Use pictures and diagrams to show features of weathering on granite in Seychelles. • List the different types of mass movements that occur on slopes: (Soil creep, landslides, mudflows, rock falls). • Describe the factors which cause each type of mass movement. • Give examples of ways in which human action can trigger or accelerate mass movements on slopes.	• Discussions. • Written activities. • Teacher explanation. • Group work. • Research/ Project activities. • Practical activities: e.g. looking for evidence of weathering and mass movements in the local area.	• The New Wider World • Physical Geography in diagrams for Africa. • Discover Physical Geography. • Geography in Place.	• Structured source-based written questions. • Informal assessments based on classroom performance.

CYCLE 4

Secondary Three - Term 2

Unit 20: Hydrology and River Studies

Duration: 27 periods of 40 minutes [9 weeks]
Split into 4 weeks Term 2 and 5 weeks Term 3

Introduction:

In this unit students will be provided with a thorough understanding of river processes and the resulting features. Students will also be able to identify the different stages of a river and different types of drainage patterns. This unit will provide them with opportunities to develop investigative skills in the field.

Terminal objectives:

1. Know what the term hydrology refers to.
2. Understand the terminologies associated with rivers.
3. Know the main types of rivers.
4. Recognise the main types of drainage patterns.
5. Understand the profile of a river.
6. Illustrate with the aid of a diagram, the long and cross profile of a river.
7. Know the processes that occur in rivers.
8. Understand how materials are transported along the course of a river.
9. Understand when and why deposition occurs in a river.
10. Recognise features formed in the upper, middle and lower course of a river.
11. Understand the effects of flooding on river processes.
12. Know the uses of rivers to humans.
13. Know the ways in which human misuse rivers.

Unit Assessment:

The terminal objectives of this unit will be assessed and graded based on:

(i) Two written assessments. One based on objectives 1 to 10 and one based on objectives 11to 13.
(ii) Different pieces of work done during the course of the unit.
(iii) Informal assessment based on knowledge, cognitive skills and attitudes.

Geography Teaching/Learning Scheme

Unit 20: Hydrology and River Studies

Enabling objectives	Teaching/Learning Strategies	Resources	Assessment
• Define the term hydrological cycle. • Describe the hydrological cycle using a diagram. • Describe the river system in terms of inputs, outputs and processes. • Describe drainage basin with the aid of a diagram. • Define all the terms associated with the drainage basin. • State that rivers can be intermittent, seasonal, or permanent. • Give examples of each of the types of rivers above, in the local area wherever possible. • Define the term drainage pattern. • List and explain the four main types of drainage patterns: dendritic, radial, trellised, centripetal. • Explain with the aid of diagrams the meaning of the long profile and cross profile of rivers. • Describe the three processes that occur in rivers, (erosion, transportation and deposition). • State the rivers can be separated into three identifiable stages: the upper, middle and lower stage. • Name and explain the four processes of erosion that occur in rivers: hydraulic action, corrosion, corrosion, attrition.	• Explanation • Discussion. • Drawing activities. • Project work • Group-work • Written activities. • Research work • Fieldwork • Map interpretation	• The Human Geography of Seychelles. • Steps in Geography Book one. • The New Wider World. • Discover Physical Geography. • Reading Tropical Map. • KS3. • Geography Complete IGCSE.	• Structured source-based written questions. • Informal assessments based on classroom performance.

• Describe the ways in which a river transports its load: traction, saltation, solution, suspension. • With the aid of diagrams, explain the formation of the main erosional features found in the upper course of the river: V-shaped valleys, gorges, interlocking spurs, rapids, waterfalls, pot holes, plunge pools. • With the aid of diagrams, explain the formation of the main features found in the middle stage of the river: meanders, river cliffs, slip off slopes. • With the aid of diagrams, explain the formation of the main features found in the lower course: floodplain, levees, ox-bow lakes, meanders, deltas. • Case Study: Bangladesh - State the conditions that give rise to floods. - Describe the effects of flooding to the human and natural environment. - List the use and misuse of rivers. - List measures/ strategies to reduce flooding (Flood management).			

CYCLE

Secondary 3 - Term 3

Unit 21: Coastal Landforms **Duration: 21 periods of 40 minutes [7 weeks]**

Introduction:

In this unit, the S3 students will be provided with a thorough understanding of the work of waves in eroding a coastline and re-sorting transporting and depositing materials removed through erosion.

The importance of rock type and structure, on coastal landforms, the action of wind on coastal sand dunes will be stressed.

The students will be able to describe the shape of the following landforms, cliffs, arches, stacks, wave cut platforms, spits bars, beaches, coastal marshes and they will be able to relate these descriptions to the processes which have led to their formation. Students will be exposed to the description of coral reefs, fringing reefs, barrier reefs and atolls and the conditions which favour their formation.

Terminal objectives:

11. Demonstrate an understanding of wave processes in eroding a coastline and re-sorting and depositing materials removed through erosion
12. Understand the type of waves (constructive and destructive) and the components of waves swash and backwash
13. Understand the different erosional processes of wave action (corrasion, hydraulic action, corrosion and attrition)
14. Understand the transportation process of materials along a coastline (long shore drift)
15. Demonstrate an understanding of the coastal landforms by describing the processes which led to their formation

Unit Assessment:

The terminal objectives of this unit will be assessed and graded based on:

(iv) written assessment based on terminal objectives 1 to 3 and 4 to 5 respectively
(v) different pieces of written work done during the course of the unit
(vi) informal oral assessment based on knowledge, cognitive skills and attitudes

Geography Teaching/Learning Scheme

Unit 21: Coastal Landforms

Enabling objectives	Teaching/Learning Strategies	Resources	Assessment
Coastal landforms • Define coast • By referring to examples in Seychelles classify coasts as highland or lowland • Mention terminologies such as high tide level, low tide level, currents etc • Mention that waves are the main agent forming features on coasts but that winds and currents also help ***Wave Action*** • Explain how waves are formed and with the help of a diagram show its structure (height. length, crest, trough) • Briefly explain refraction of waves in shallow water and the breaking of waves (swash and backwash) • Mention that waves perform 3 main functions (erosion, transportation and deposition) bringing out the fact that for any given stretch of coast, at any given time, erosion balances deposition • Note that waves carve on features along highland coasts by erosion and build new features on lowland coasts by deposition ***Wave Erosion*** • Mention that the rate of erosion depends on rock resistance.	• Explanation • Discussion • Drawing activities • Project work • Open pair work • Written activities • Research work • Field work • Map interpretation	• The New Wider World • Discover Physical Geography • Steps in Geography • KS3. • Geography Complete IGCSE. • Geography Course book Sibley. • Videos.	• Structure source based written question • Informal assessment based on classroom performance •

Geography Teaching/Learning Scheme

Unit 21: Coastal Landforms

Enabling objectives	Teaching/Learning Strategies	Resources	Assessment
• List and explain the processes of wave erosion (hydraulic action, corrosion, attrition and solution) • With the help of diagrams list and explain the main features of erosion found along the coast stating where they are found exactly (blow hole, cliffs, above high tide level) headland, caves, arches, stacks, wave cut platform, below high tide level, and stumps at low tide level. Use examples. • With the help of diagrams show the formation of bays giving local examples ***Transportation of materials*** • With reference to a diagram show how movement of materials takes place long the coast • List the 3 agents which move material (mainly waves, long shore currents, currents and wind) ***Deposition by Waves*** • With the help of a diagram show how waves sort out and grade their materials before depositing them			

Enabling objectives	Teaching/Learning Strategies	Resources	Assessment
• With reference to diagrams and examples list the main depositional features found along the coast (beaches, bars and lagoons, spits, tombolos and coastal marches or swamps) Give examples of places. • Show how human activities can interfere with the movement of materials along coasts (groynes). • Mention that sand dunes along the coast are a result of wind deposition. • Case Study: Coastal Management: Opportunities, Hazards and Management. ***Other coastal features coral reefs ecosystem.*** • Explain how coral is formed and list the necessary conditions for its growth temperature, depth, sunlight, clear fresh water) • Name and draw the 3 types of coral formations (fringing reefs, barrier reefs and atolls) • Explain the three types of coral formation • Mention that they are features found only on some tropical coasts • Give examples of some of these formations in Seychelles and elsewhere (e.g. Great Barrier Reef) • Mention the importance of coral reefs to the economy of a country (refer to Seychelles as an example) • Mention that coral reefs form part of a very fragile marine ecosystem. • Mention how the reefs of Seychelles are protected.			

CYCLE 5 (IGCSE PROGRAMME)

Theme 1: Population and settlement

1.1 Population dynamics

Candidates should be able to:

Describe and give reasons for the rapid increase in the world's population
Show an understanding of over-population and under-population
Understand the main causes of a change in population size
Give reasons for contrasting rates of natural population change
Describe and evaluate population policies

Further guidance

Causes and consequences of over-population and under-population

How birth rate, death rate and migration contribute to the population of a country increasing or declining

Impacts of social, economic and other factors (including government policies, HIV/AIDS) on birth and death rates

Case Studies required for 1.1

- A country which is over-populated
- A country which is under-populated
- A country with a high rate of natural population growth
- A country with a low rate of population growth (or population decline)

1.2 Migration

Candidates should be able to:

Explain and give reasons for population migration

Demonstrate an understanding of the impacts of migration

Further guidance

Internal movements such as rural-urban migration, as well as international migrations, both voluntary and involuntary

Positive and negative impacts on the destination and origin of the migrants, and on the migrants themselves

Case Study required for 1.2

- An international migration

1.3 Population structure

Candidates should be able to:

Identify and give reasons for and implications of different types of population structure

Case Study required for 1.3

- A country with a high dependent population

Further guidance

Age/sex pyramids of countries at different levels of economic development

1.4 Population density and distribution

Candidates should be able to:

Describe the factors influencing the density and distribution of population

Case Studies required for 1.4

Physical, economic, social and political factors

Further guidance

- A densely populated country or area (at any scale from local to regional)
- A sparsely populated country or area (at any scale from local to regional)

1.5 Settlements (rural and urban) and service provision

Candidates should be able to:

Explain the patterns of settlement

Describe and explain the factors which may influence the sites, growth and functions of settlements

Give reasons for the hierarchy of settlements and services

Case Study required for 1.5

Settlement and service provision in an area

Further guidance

Dispersed, linear, and nucleated settlement patterns Influence of physical factors (including relief, soil, water supply) and other factors (including accessibility, resources)

High-, middle- and low-order settlements and services. Sphere of influence and threshold population

1.6 Urban settlements

Candidates should be able to:

Describe and give reasons for the characteristics of, and changes in, land use in urban areas

Explain the problems of urban areas, their causes and possible solutions

Case Study required for 1.6

- An urban area or urban areas

Further guidance

Land use zones including the Central Business District (CBD), residential areas, industrial areas and the rural-urban fringe of urban areas in countries at different levels of economic development

The effect of change in land use and rapid urban growth in an urban area including the effects of urban sprawl

Different types of pollution (air, noise, water, visual), inequality, housing issues, traffic congestion and conflicts over land use change

1.7 Urbanisation

Candidates should be able to:

Identify and suggest reasons for rapid urban growth

Describe the impacts of urban growth on both rural and urban areas, along with possible solutions to reduce the negative impacts

Case Study required for 1.7

A rapidly growing urban area in a developing country and migration to it

Further guidance

Reference should be made to physical, economic and social factors which result in rural depopulation and the movement of people to major cities

The effects of urbanisation on the people and the natural environment

The characteristics of squatter settlements Strategies to reduce the negative impacts of urbanisation

Theme 2: The natural environment

2.1 Earthquakes and volcanoes

Candidates should be able to:

Describe the main types and features of volcanoes and earthquakes

Describe and explain the distribution of earthquakes and volcanoes

Describe the causes of earthquakes and volcanic eruptions and their effects on people and the environment

Demonstrate an understanding that volcanoes present hazards and offer opportunities for people

Explain what can be done to reduce the impacts of earthquakes and volcanoes

Further guidance
Types of volcanoes (including strato-volcanoes [composite cone] and shield volcano)
Features of volcanoes (including crater, vent, magma chamber)
Features of earthquakes (including epicentre, focus, magnitude)
The global pattern of plates and their structure; an awareness of plate movements, subduction zones and their effects – constructive/divergent, destructive/convergent and conservative plate boundaries

Case Studies required for 2.1

- An earthquake
- A volcano

2.2 Rivers
Candidates should be able to:
Explain the main hydrological characteristics and processes which operate in rivers and drainage basins
Demonstrate an understanding of the work of a river in eroding, transporting and depositing
Describe and explain the formation of the landforms associated with these processes

Demonstrate an understanding that rivers present hazards and offer opportunities for people

Explain what can be done to manage the impacts of river flooding

Case Study required for 2.2
The opportunities presented by a river or rivers, the associated hazards and their management

Further guidance
Characteristics of rivers (including width, depth, speed of flow, discharge) and drainage basins (including watershed, tributary, confluence)
Processes which operate in a drainage basin (including interception, infiltration, throughflow, groundwater flow, evaporation, overland flow)

Forms of river valleys – long profile and shape in cross-section, waterfalls, potholes, meanders, oxbow lakes, deltas, levées and flood plains
Causes of hazards (including flooding and river erosion)
Opportunities of living on a flood plain or a delta or near a river

2.3 Coasts
Candidates should be able to:
Demonstrate an understanding of the work of the sea and wind in eroding, transporting and depositing
Describe and explain the formation of the landforms associated with these processes

Describe coral reefs and mangrove swamps and the conditions required for their development

Demonstrate an understanding that coasts present hazards and offer opportunities for people
Explain what can be done to manage the impacts of coastal erosion

Case Study required for 2.3

The opportunities presented by an area or areas of coastline, the associated hazards and their management

Further guidance

Cliffs, wave-cut platforms, caves, arches, stacks, stumps, bay and headland coastlines, beaches, spits, and coastal sand dunes

Hazards (including coastal erosion and tropical storms)

2.4 Weather

Candidates should be able to:
Describe how weather data are collected
Make calculations using information from weather instruments
Use and interpret graphs and other diagrams showing weather and climate data

Further guidance
Describe and explain the characteristics, siting and use made of a Stevenson Screen
Rain gauge, maximum-minimum thermometer, wet-and-dry bulb thermometer (hygrometer),
sunshine recorder, barometer, anemometer and wind vane, along with simple digital instruments which can be used for weather observations; observations of types and amounts of cloud

2.5 Climate and natural vegetation

Candidates should be able to:
Describe and explain the characteristics of two climates:
- equatorial
- hot desert

Describe and explain the characteristics of tropical rainforest and hot desert ecosystems
Describe the causes and effects of deforestation of tropical rainforest

Case Study required for 2.5

- An area of tropical rainforest
- An area of hot desert

Further guidance

Climate characteristics (including temperature [mean temperature of the hottest month, mean temperature of the coolest month, annual range]; and precipitation including convection and relief rainfall [the amount and seasonal distribution])

Factors influencing the characteristics of these climates (including latitude, pressure systems, winds, distance from the sea, altitude and ocean currents)

Climatic graphs showing the main characteristics of temperature and rainfall of the two climates
The relationship in each ecosystem of natural vegetation, soil, wildlife and climate
Effects on the natural environment (both locally and globally) and effects on people

Theme 3: Economic development

3.1 Development

Candidates should be able to:
Use a variety of indicators to assess the level of development of a country

Identify and explain inequalities between and within countries
Classify production into different sectors and give illustrations of each
Describe and explain how the proportions employed in each sector vary according to the level of development
Describe and explain the process of globalisation, and consider its impacts

Case Study required for 3.1

- A transnational corporation and its global links

Further guidance
Indicators of development (including GNP per capita, literacy, life expectancy and composite indices, e.g. Human Development Index (HDI))

Primary, secondary, tertiary and quaternary sectors Use of indicators of development and employment
structure to compare countries at different levels of economic development and over time
The role of technology and transnational corporations in globalisation along with economic factors which give rise to globalisation

Impacts at a local, national and global scale

3.2 Food production

Candidates should be able to:
Describe and explain the main features of an agricultural system: inputs, processes and outputs
Recognise the causes and effects of food shortages and describe possible solutions to this problem

Further guidance
Farming types: commercial and subsistence; arable, pastoral and mixed; intensive and extensive
The influence of natural and human inputs on agricultural land use (including natural inputs [relief, climate and soil] and human inputs [economic and social]). Their combined influences on the scale
of production, methods of organisation and the products of agricultural systems

Natural problems which cause food shortages (including drought, floods, tropical storms, pests) and economic and political factors (including low capital investment, poor distribution/transport difficulties, wars)

The negative effects of food shortages; the effects of food shortages in encouraging food aid and measures to increase output

Case Studies required for 3.2

- A farm or agricultural system
- A country or region suffering from food shortages

3.3 Industry

Candidates should be able to:

Demonstrate an understanding of an industrial system: inputs, processes and outputs (products and waste)

Describe and explain the factors influencing the distribution and location of factories and industrial zones

Case Study required for 3.3

- An industrial zone or factory

Further guidance

Industry types: manufacturing, processing, assembly and high technology industry

The influence of factors including land, labour, raw materials and fuel and power, transport, markets and political factors

Their combined influences on the location, scale of production, methods of organisation and the products of the system

Industrial zones and/or factories with respect to locational and siting factors

3.4 Tourism

Candidates should be able to:

Describe and explain the growth of tourism in relation to the main attractions of the physical and human landscape

Evaluate the benefits and disadvantages of tourism to receiving areas

Demonstrate an understanding that careful management of tourism is required in order for it to be sustainable

Case Study required for 3.4

- An area where tourism is important

3.5 Energy

Candidates should be able to:

Describe the importance of non-renewable fossil fuels, renewable energy supplies, nuclear power and fuelwood; globally and in different countries at different levels of development

Evaluate the benefits and disadvantages of nuclear power and renewable energy sources

Case Study required for 3.5

- Energy supply in a country or area

Further guidance
Non-renewable fossil fuels including coal, oil and natural gas. Renewable energy supplies including geothermal, wind, HEP, wave and tidal power, solar power and biofuels

3.6 Water

Candidates should be able to:

Describe methods of water supply and the proportions of water used for agriculture, domestic and industrial purposes in countries at different levels of economic development

Explain why there are water shortages in some areas and demonstrate that careful management is required to ensure future supplies

Case Study required for 3.6

- Water supply in a country or area

Further guidance

Methods of water supply (including reservoirs/dams, wells and bore holes, desalination)

The impact of lack of access to clean water on local people and the potential for economic development

3.7 Environmental risks of economic development

Candidates should be able to:
Describe how economic activities may pose threats to the natural environment and people, locally and globally
Demonstrate the need for sustainable development and management

Understand the importance of resource conservation

Further guidance
Threats to the natural environment (including soil erosion, desertification, enhanced global warming and pollution [water, air, noise, visual])
Case Study required for 3.7

- An area where economic development is taking place and causing the environment to be at risk

Using mathematical skills in geography

Candidates should be able to:

- add, subtract, multiply and divide

- understand map scale and the use of the scale line and representative fractions
- understand the terms *mean* and *range*
- use averages, decimals, fractions, percentages and ratios
- use standard notation, including both positive and negative indices
- understand significant figures and use them appropriately
- recognise positive and negative relationships shown by scatter graphs
- draw and interpret graphs from given data for the types of graph listed in the syllabus
- select suitable scales and axes for graphs
- use a ruler and protractor.

Calculators may be used in all examinations.